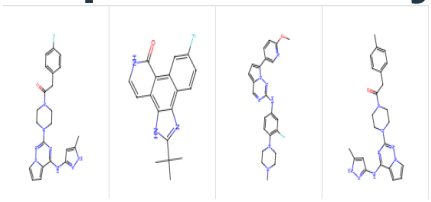
(a) Inputs

Compound library



ECFP4 (2048)

Primary assays
t_0 observed so far

Historical assays
t_1–t_3 historical

(b) GP strategies

STL Single-task
t_0 only

MT Multi-task
t_0 + t_1–t_3 | fit r

Pooled
t_0 + t_1–t_3 | r = 1

Target-wise z-scoring

(c) Sequential screening

Posterior for t_0
μ_0, σ_0 per policy

Expected Improvement
best observed t_0

Select next batch

Assay primary target t_0

Add t_0 assays & refit after each batch